STRhub Verified

strhub.app · strhub.app/verified

Technical Reproducibility Validation Report

Strspy

Version v2.0
Autosomal STR (ONT)

Verification date	2026-09-01
Repository commit	dafdee7e7e5672c8dc732e8577dbe153f53a12f5
Attestation level	**Runs + Expected IO**
Permalink	https://strhub.app/verified/strspy-v2-0-ont
STRhub	https://strhub.app

SCOPE OF THIS REPORT

Independent automated verification of reproducible execution. It confirms the tool installs, runs end-to-end, and produces structurally valid output in a standardized environment.

This is **not** an analytical validation. Genotype accuracy, concordance, forensic suitability, and regulatory compliance are out of scope.

WHAT WAS VERIFIED

– Source available
– Installation successful
– End-to-end execution
– Output generated

NOT VERIFIED

– Genotype accuracy
– Concordance
– Forensic validity
– Regulatory compliance

1. Executive Summary

PURPOSE	Verify that the tool installs, runs end-to-end, and produces structurally valid output.
RESULT	**4/5 gates passed. Runs + Expected IO**
ATTESTATION LEVEL	**Runs + Expected IO**
SCOPE	Installation, execution, and output structure verification.
NOT EVALUATED	Genotype accuracy · Concordance · Forensic validity · Regulatory compliance

2. Reproducibility Metadata

Tool	Strspy
Version	v2.0
Repository	https://github.com/unique379r/strspy
Commit (pinned)	dafdee7e7e5672c8dc732e8577dbe153f53a12f5
Container OS	ubuntu-22.04
Marker panel	Autosomal STR (ONT)
Verified on	2026-09-01 11:18:36 UTC
CI run	https://github.com/Tfronta/strhub-verified/actions/runs/33501209177
Submitted by	A third party (not the tool's maintainer)

This tool was submitted for verification by somebody other than its maintainer. The maintainer took no part in the run and supplied none of what it used: the command, the environment, and any target regions were chosen by the submitter. This certificate records a run; it is not an endorsement by the tool's maintainer, and does not imply their involvement.

3. Exact Run Command

This is the command that was executed, verbatim, in the CI environment.

```
# Environment: ubuntu-22.04 · Strspy v2.0 · commit dafdee7
$ bash /opt/strspy/src/STRspy_Normal_v2.0_Args.sh \
-s /data/in \
-r yes \
-t ont \
-f /opt/strspy/db/STRspy2.0-DB \
-b /opt/strspy/db/STRspy2.0-DB \
-l /opt/strspy/db-v2/STRspy_v2.DB.sort.bed \
-k 0.4 \
-o /data/out \
-c /opt/strspy/config/ToolsConfig.txt \
-d 1
```

4. Verification Gates

Available	The pinned public source exists	PASS
Installs	The environment builds from source	PASS
Runs	Executes end-to-end without crashing	PASS
Expected IO	Produces a non-empty file in the declared format	PASS
Plausible Output	Output structure matches expected genotype-bearing format	—
Execution log	strspy-v2-0-ont.log-external.txt	

Stopping at a step is not a finding that the software is faulty. It records how far this particular attempt got. Software can stop early because a dependency it names is no longer available, because it expects input arranged differently from the reference sample, or because the automated environment cannot supply something it needs, such as commercially licensed components. The messages on screen when it stopped are reported below.

5. Out of Scope

This report does not evaluate any of the following:

– Genotype correctness or accuracy
– Concordance against known truth sets
– Sensitivity, specificity, or stutter performance
– Allele calling accuracy or forensic casework suitability
– Regulatory compliance or ISO accreditation
– Multi-laboratory or multi-dataset reproducibility

6. Output Content Evidence

Output file	**/*.txt
Format	TSV
Sequence records	Not assessed
Distinct STR loci	Not assessed
Total reads	Not assessed
Max depth (single locus)	Not assessed

7. Verification Data

This tool does not include its own demo or test data. STRhub ran the verification using the public reference dataset listed below. A small test file in the repository lets a new user run the tool on their first day and see it working before trusting it with their own data, and it lets a verification run against the author's own sample as well as this one. Publishing the output that file should produce helps just as much: it shows what the results are meant to look like, which is what a reader needs to tell a correct run from one that merely finished.

Dataset	1000 Genomes ONT, hg38 CODIS slice (R10 SUP)
Source	https://s3.amazonaws.com/1000g-ont/index.html?prefix=PROCESS…
License	Open access (1000 Genomes / HPRC). Research use. STRhub is not the data provider.
Ref. genome	GRCh38 / hg38
Loci tested	—

8. Verification Matrix

PASS	Reference dataset	1000 Genomes ONT, hg38 CODIS slice (R10 SUP)
	README check (5/5)	Advisory: all items present
	Install / environment setup	Present
	Run command	Present
	Expected input format	Present
	Produced output format	Present
	Dependencies / versions	Present

9. Errors Reported During the Run

The tool reported errors on 9 item(s) during the run. This does not assess whether the results produced are correct.

What happened	Times	Affected
Could not open expected files	18	D10S1248, D12S391, D13S317, D5S818, D7S820, D8S1179, FGA, TPOX, vWA

Structural errors, such as a file that will not open, an unrecognized command-line flag, or an incomplete build, do not depend on the sample: a coverage-limited slice yields fewer reads, but it cannot cause them. These are not attributable to STRhub's reference sample.

A small test file in the tool's own repository lets a new user run it on their first day and see it working before trusting it with their own data, and it lets a verification run against the author's sample as well as STRhub's slice. Publishing the output that file should produce helps just as much: it shows what the results are meant to look like, which is what a reader needs to tell a correct run from one that merely finished.

10. What This Run Needed Beyond the Repository

The result above describes a run configured as follows. Anyone repeating it needs the same things.

– Test data: no sample from the repository was used, so a public reference sample stood in.

11. Limitations

– Single reference dataset per input type
– Single containerized environment (Docker / ubuntu-22.04)
– No truth-set comparison or ground-truth genotypes
– No accuracy or concordance assessment
– No forensic validation of results
– Short-read limitations apply (very long STR alleles may not span reads)

12. Scope and Disclaimers

Executed end-to-end in the stated environment with output in the expected format. Concerns reproducible execution only; no claim of accuracy, casework fitness, or regulatory validation.

Each result is a dated snapshot, verified on the tool's public repository at a pinned commit. STRhub does not store tool source code.

13. Conclusion

◆ **Runs end-to-end**

Strspy v2.0 installs and executes without error in a clean ubuntu-22.04 environment at the pinned commit. All 4 verification gates were passed.

◆ **Produces valid output**

The tool generated a structurally valid TSV output file with genotype calls across 0 target forensic STR loci, with a maximum read depth of 0 at —.

◆ **No bundled demo data**

Strspy does not include its own test or demo dataset. STRhub Verified supplied 1000 Genomes ONT, hg38 CODIS slice (R10 SUP) as the reference input for this verification run. Users replicating this result should use the same or equivalent publicly available reference material.

◆ **Reproducibility statement**

The exact command reported in Section 3, executed at the pinned commit in the described environment, is sufficient to reproduce this verification result independently.